

**Report:** b938f7cf  
**Generated:** 2026-07-31 04:33:27Z  
**Run started:** not recorded  
**Software:** OriginMarker 2.2.1 (Kinetochore), Syngamy module  
**Sperm donor:** GSM4472397\_sperm\_DNA\_71.CEL.txt.gz (825,656 markers, SHA-256 1b56911dae7030f4)  
**Oocyte donor:** none supplied, so a maternal origin is inferred from paternal absence rather than measured  
**Donor heterozygosity:** 16.658% autosomal, of called markers  
**Samples:** 1 analysed (1 androgenetic)  
**Assembly:** GRCh37  
**Data handling:** every file was read in the browser that produced this report. No genotype was transmitted, and none is retained.

## Result summary

| Sample                          | Origin       | Absence | Ceiling | x ceiling | Zygosity               | Sperm     | Informative |
|---------------------------------|--------------|---------|---------|-----------|------------------------|-----------|-------------|
| GSM4774683_pMII-4_56.CEL.txt.gz | androgenetic | 0.57%   | 0.93%   | 0.6x      | uniparental homozygous | X-bearing | 547,115     |

Absence is the rate at which the donor's obligate allele is missing where he is homozygous. Ceiling is what this sample's own noise can manufacture: its no-call rate times its heterozygous fraction. A sample is called as lacking a paternal contribution only above 3x that ceiling, and as carrying one only at or below it; in between it is left uncalled. No confidence percentage is reported, deliberately: see Interpretation.

GSM4774683\_pMII-4\_56.CEL.txt.gz vs GSM4472397\_sperm\_DNA\_71.CEL.txt.gz

## ANDROGENETIC

Paternal genome only. No maternal complement.

absent 0.57% | ceiling 0.93% | 0.6x | 547,115 informative markers | zygosity uniparental homozygous | X-bearing sperm

## Evidence

| Measurement             | Observed | Reference | The reference is                                                                                                                                                                                                               |
|-------------------------|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Donor alleles absent    | 0.57%    | 0.93%     | this sample's no-call rate times its heterozygous fraction. Dropout manufactures absence only by turning a heterozygous call homozygous and discarding the donor's allele, so the bound is the product and neither term alone. |
| Alleles the donor lacks | 6.25%    | 8.33%     | half the donor's heterozygosity. A second parent supplies an allele he lacks only where he is homozygous, at sum(pq), and his heterozygosity is sum(2pq).                                                                      |
| Heterozygous BAF band   | 2.85%    | 8.00%     | measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.                                                                   |

## Sex chromosomes

| Chr  | Informative | Absent | Rate  | x ceiling | Verdict | Note |
|------|-------------|--------|-------|-----------|---------|------|
| chrX | 16,906      | 192    | 1.14% | 1.2       | unclear |      |

## Chromosomes

| Chr   | Informative | Absent | Rate  | x ceiling | Verdict | Note |
|-------|-------------|--------|-------|-----------|---------|------|
| chr1  | 45,200      | 268    | 0.59% | 0.6       | present |      |
| chr2  | 42,596      | 184    | 0.43% | 0.5       | present |      |
| chr3  | 36,670      | 164    | 0.45% | 0.5       | present |      |
| chr4  | 33,133      | 159    | 0.48% | 0.5       | present |      |
| chr5  | 32,025      | 135    | 0.42% | 0.5       | present |      |
| chr6  | 38,373      | 142    | 0.37% | 0.4       | present |      |
| chr7  | 29,134      | 177    | 0.61% | 0.7       | present |      |
| chr8  | 27,021      | 146    | 0.54% | 0.6       | present |      |
| chr9  | 23,584      | 149    | 0.63% | 0.7       | present |      |
| chrX  | 16,906      | 192    | 1.14% | 1.2       | unclear |      |
| chr10 | 26,787      | 130    | 0.49% | 0.5       | present |      |
| chr11 | 29,114      | 188    | 0.65% | 0.7       | present |      |
| chr12 | 26,574      | 99     | 0.37% | 0.4       | present |      |
| chr13 | 18,145      | 150    | 0.83% | 0.9       | present |      |
| chr14 | 18,600      | 106    | 0.57% | 0.6       | present |      |
| chr15 | 17,097      | 54     | 0.32% | 0.3       | present |      |
| chr16 | 19,549      | 270    | 1.38% | 1.5       | unclear |      |
| chr17 | 19,962      | 162    | 0.81% | 0.9       | present |      |
| chr18 | 15,294      | 48     | 0.31% | 0.3       | present |      |
| chr19 | 18,840      | 178    | 0.94% | 1.0       | unclear |      |
| chr20 | 13,209      | 94     | 0.71% | 0.8       | present |      |
| chr21 | 7,754       | 60     | 0.77% | 0.8       | present |      |
| chr22 | 8,454       | 70     | 0.83% | 0.9       | present |      |

## Sample quality

**Markers:** 825,656 read, 701,745 called (85.0%)  
**No-call:** 15.01%

**Heterozygous:** 7.60% of called  
**chrX / autosomal het:** 0.831  
**Sex call:** female  
**Product:** UK Biobank Axiom Array  
**Assembly:** GRCh37 (0 placements illegal under GRCh37, 6,315 under GRCh38, 825,656 tested)  
**Allele coding:** ok BAF corroborates 0=AA, 2=BB  
**Mean BAF AA/AB/BB:** 0.043 / 0.542 / 0.966  
**SHA-256:** 8a8cbfcc1b87e1330107d4e74fc9618305aa4f64e31d3cd82b15dc204de05b46  
**File size:** 63,529,038 bytes

Quality gates

| Gate                       | Value | Verdict     | Basis                                                                                                                                                 |
|----------------------------|-------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| call rate                  | 85.0% | usable      | inside the 75-95% amplified usable band; the Axiom vendor gate of 97% would reject this sample, and that gate was established on unamplified bulk DNA |
| het-to-hom asymmetry valid | 85.0% | usable      | holds within the usable call-rate band                                                                                                                |
| genome-wide LOH            | 7.6%  | report only | no genome-wide LOH signature                                                                                                                          |
| heterozygosity plausible   | 7.6%  | usable      | within the plausible range for one diploid genome                                                                                                     |
| numeric genotype coding    | -     | usable      | BAF corroborates 0=AA, 2=BB                                                                                                                           |
| sex call                   | 83.1% | usable      | chrX/autosome heterozygosity ratio 0.831 is inside the female band. Dropout cancels in the ratio, so this survives amplified material.                |

Findings

Every allele traces to this parent and the genome is homozygous: a parent-only complement, duplicated. Nothing here required chrY. No chrY, yet the parent's alleles are present on chrX as well as the autosomes: an X-bearing sperm. This is the case a chrY test cannot call.

Methods

Parent of origin was determined from SNP array genotypes using OriginMarker 2.2.1 (Kinetochore). For each of 1 sample(s), the rate at which the sperm donor's obligate allele was absent was computed across autosomal markers where he is homozygous and the sample is called, and compared against the rate that sample's own genotyping noise can produce, taken as the product of its no-call rate and its heterozygous fraction. A sample was called as lacking a paternal contribution only where the observed rate exceeded that bound by 3-fold, and as carrying one only where it fell at or below the bound; rates between the two were left uncalled. Whether a second parent contributed was assessed from the rate at which the sample carries alleles the sperm donor does not possess, against half his heterozygosity, which is the contribution a second parent makes. Zygosity was taken from the fraction of B-allele frequencies falling in the heterozygous band, 0.35 to 0.65. Sperm type was inferred from the chrX rate rather than from chrY, so that an X-bearing sperm carrying a full paternal genome is distinguishable from an absent paternal contribution; neither case leaves a Y.

No oocyte donor array was supplied. A maternal origin below is therefore inferred from the absence of the paternal contribution rather than measured, and a sample belonging to neither declared parent is indistinguishable from one that is maternal in origin.

Assembly was determined from marker positions against the UCSC chromInfo and gap tables and found to be GRCh37. No liftOver was performed.

Interpretation

No confidence percentage is attached to a call, and this is deliberate. The per-sample likelihood ratio between "paternal genome present" and "absent" runs past 10^10000 on a genome-wide array, which is not a probability any reader should be handed. The honest figure is the empirical accuracy of the method, and on the pairs of known relationship it has been run against that is 9 of 9 correct, whose 95% lower bound is roughly 72%. About 300 consecutive correct calls would be needed to claim 99%. What is reported instead is the margin: the observed rate, the reference it was measured against, and the ratio between them, so a reader can see how far from the boundary the call sits.

The absence axis separates by roughly thirty-fold and carries the call. The alleles-the-donor-lacks axis separates by about 1.6-fold and only distinguishes a biparental genome from a paternal-only one that is heterozygous; treat a near-boundary call on that axis as provisional. A sample scoring in the uncalled band is not a weak positive, it is an absence of evidence: an array from the second parent measures dropout directly and settles it.

Constants and where they come from

| Constant                      | Value                                 | Provenance                                                                                                                                                       |
|-------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mendelian inconsistency floor | 0.0063                                | Kothiyal 2019, 0.63% of variant sites across 1,314 nuclear families. A lower bound on the spurious-violation rate, never the value itself.                       |
| Paternal-absence noise bound  | no-call rate x heterozygous fraction  | Derived per sample, not fitted. Bounds every related pair measured, within 2x: 0.13% predicted against 0.05% observed, 0.18% against 0.16%, 10.4% against 9.69%. |
| Absence calling margin        | 3x                                    | How far above the noise bound an absence must sit before it is called.                                                                                           |
| Residual absence floor        | 0.005                                 | Absence on clean data from genotyping error alone, measured at 0.03% and 0.05%. Added to the bound so a near-perfect array still has a non-zero reference.       |
| Second-parent signal          | half the donor's heterozygosity       | Derived. A second parent supplies an allele the donor lacks only where he is homozygous, at sum(pq); his heterozygosity is sum(2pq) over the same markers.       |
| Diploid heterozygous BAF band | 0.08 of markers in 0.35-0.65          | Measured at 15-16% for diploid genomes and 1.3-3.4% for uniparental ones. Dropout does not mimic this: a biparental embryo at 33% dropout still reads 22.2%.     |
| Per-chromosome minimum        | 200 informative markers               | Below this a chromosome is not reported: the rate is too noisy to place against the ceiling.                                                                     |
| Assembly tables               | UCSC chromInfo and gap, hg19 and hg38 | Freely redistributable. No liftOver chain file is used or shipped: those carry a non-commercial field-of-use restriction Apache 2.0 cannot sublicense.           |

Files read

| File                               | Role   | Bytes      | Markers | SHA-256                                                          |
|------------------------------------|--------|------------|---------|------------------------------------------------------------------|
| GSM4472397_sperm_DNA_71.CEL.txt.gz | donor  | 63,674,062 | 825,656 | 1b56911dae7030f4ecec2ade9c7b1bacee23def81d130e6b3d0ac9edb388169  |
| GSM4774683_pMII-4_56.CEL.txt.gz    | sample | 63,529,038 | 825,656 | 8a8cbfcc1b87e1330107d4e74fc9618305aa4f64e31d3cd82b15dc204de05b46 |

The hash is of the file as read, so a reviewer can confirm the input without being sent it.

Citation

OriginMarker 2.2.1 (Kinetochore). Syngamy: parent of origin from SNP array genotypes. Report b938f7cf, generated 2026-07-31 04:33:27Z. Kothiyal P et al. Mendelian inconsistent signatures from 1,314 ancestrally diverse family trios. J Comput Biol 2019;26(5):405-419. doi:10.1089/cmb.2018.0253. Cited for the inconsistency floor above.